

YOUSSEF ELSOKKARY

Computing and Economics Student

joe.sokkary@gmail.com

4165800136

GitHub: <https://github.com/youssef061204>

Portfolio; Personal Website: <https://youssefelsokkary.vercel.app/>

EDUCATION & CERTIFICATIONS; QUEEN'S UNIVERSITY

- **Honours Bachelor's Degree in Computing; Expected 2028**
- **Double Major: Artificial Intelligence and Finance & Data Analysis**
- Relevant Coursework: Intro to CS I & II Discrete Structures I & II Calculus I & II, Statistics, **Linear methods for AI**, Logic in Computing, **Data Structures and Algorithms**, Econometrics, Computer Architecture, Software Architecture

PROJECTS

- **LedgerMatch — ML-Assisted Payment Reconciliation Platform**
TypeScript, Python, Next.js, Node.js, PostgreSQL, Prisma, XGBoost, scikit-learn, Pandas, Zod, Docker, Playwright, Vitest
 - Improved payment-to-invoice ranking accuracy from **24.13% to 87.15%** across 4,794 held-out synthetic exceptions by training an **XGBoost** ranker on 27 reference, amount, customer, and timing features.
 - Achieved **98.92%** recommendation precision at **68.83%** coverage across 5,759 held-out synthetic exceptions by calibrating confidence thresholds and implementing abstention with human approval.
 - Reduced median reconciliation runtime by **45.42%**, from **33.88s to 18.49s** across **100K** synthetic payments, by optimizing PostgreSQL transaction batching while preserving accounting constraints.
 - Built a reproducible evaluation pipeline with leakage-safe train/validation/test splits, robustness testing, Python-to-TypeScript model inference, and automated benchmarking.
- **AI Operator — Secure Autonomous Software Engineering Agent**
TypeScript, Next.js, Node.js, Gemini API, Docker, Git, SQLite
 - Achieved **85%** autonomous coding success across 50 frozen tasks using Gemini 3.8 Flash, building a bounded agentic tool loop with schema-validated actions and independent, no-network Docker verification.
 - Prevented unsafe outcomes across **42/42** deterministic scenarios with 0 permission violations and **100%** rollback correctness by implementing digest-bound approvals, isolated Git worktrees, hash-guarded multi-file changesets, and resource-limited execution.
 - Validated **50/50** benchmark fixtures and 32 automated tests with 0 failures, integrating hidden graders, browser E2E testing, CI verification, and dependency auditing across 527 packages with 0 advisories.
- **TradePersona — Behavioral ML Analytics Platform**
Python, scikit-learn, Pandas, NumPy, Next.js, Node.js, Docker, Snowflake, Playwright
 - Achieved **0.779 macro F1 [95% CI: 0.735–0.820]** on a held-out benchmark of 2,000 independent synthetic trading trajectories and **224K+** trades by engineering a leakage-resistant 20-feature behavioral ML pipeline with isolated train, calibration, policy, test, and distribution-shift partitions.
 - Improved retained classification performance to **0.808** macro F1 at **85%** coverage by implementing calibrated multinomial logistic regression, confidence-based abstention, probability-sensitivity explanations, and behavioral counterfactuals for uncertain predictions.
 - Built a reproducible full-stack ML evaluation platform with versioned model/data artifacts, 52 automated tests across Python, backend, and browser E2E, Dockerized inference, CI/CD, dependency auditing, and exact experiment reproduction.

EXPERIENCE

Founder& President— Queen's AI Automation Club (QAAC) (Non-Ratified)

- Queen's University — 2025–Present
- Founded a student club focused on building AI-driven workflow automations using n8n, LLMs, and APIs to solve real student problems across campus.
- Developing semester-long projects including internship application automation, AI study assistants, and campus event aggregation systems.
- Designing and leading beginner-friendly workshops teaching n8n, prompt engineering, and practical automation skills.
- Recruiting and managing a 5-person executive team; responsible for club operations, outreach, and ratification preparation.

Software Engineer — Merlin Neurotech Club

- Developed and optimized software linking EMG sensors to a robotic exoskeleton, enabling real-time leg motion based on muscle activation.
- Implemented signal processing algorithms and embedded system code to process sensor data with low latency(<30ms), improving responsiveness and reliability.
- Collaborated in a cross-disciplinary team to integrate hardware and software, testing and debugging complex sensor-actuator systems.

TECHNICAL SKILLS:

- Languages: Python, TypeScript, Java, JavaScript, SQL, Bash
- AI/ML: PyTorch, TensorFlow, scikit-learn, XGBoost, Pandas, NumPy, Machine Learning, Deep Learning, NLP, LLMs, AI Agents, Model Evaluation, Feature Engineering
- Backend/Web: Next.js, React, Node.js, Express, Spring Boot, REST APIs, WebSockets, Microservices
- Data/Databases: PostgreSQL, MySQL, SQLite, Snowflake, Supabase, Prisma, Data Pipelines, ETL
- Cloud/DevOps: Docker, Git, GitHub Actions, CI/CD, Linux, Vercel, Shell Scripting
- Engineering: Data Structures & Algorithms, Object-Oriented Programming, System Design, Distributed Systems, Software Testing, Playwright, Vitest, Zod, Embedded Systems